
Mathematical Explanation of the NSIS Actuarial Scenario Calculator

Four Policy Models and Their Projection Functions

This document explains only the mathematical functions implemented in the website.
It does not introduce external assumptions, laws, rates, or evidence.

Source files examined: `index.html` and `script.js`

Prepared as a VS Code/Overleaf-compatible $\text{\LaTeX}$ project

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1 What the Calculator Is

The webpage implements a **deterministic, aggregate scenario-costing model**. For each policy branch, it projects:

- the number of covered workers and their average wage;
- the expected number of claims or beneficiaries;
- the annual benefit cost, including an administrative margin;
- contribution income as a percentage of payroll;
- the accumulated reserve after investment return, contributions, and costs; and
- a level required premium rate that produces a non-negative terminal reserve over the selected projection horizon.

The four policy branches are:

1. Employment Injury Insurance;
2. Maternity Benefit Insurance;
3. Unemployment Insurance with employment-service cost; and
4. Old-Age Social Insurance.

How to read the model

The calculator does not estimate probabilities from data. The user supplies the claim rates, benefit parameters, growth rates, contribution rates, administrative margin, and fund return. The model then applies those assumptions consistently over time.

1.1 Core notation

Let $t = 0, 1, \dots, T - 1$ denote projection years, where $t = 0$ is the selected start year.

Symbol	Meaning
N_t	Number of covered workers in year t .
W_t	Average monthly wage per covered worker in year t .
P_t	Annual covered payroll in year t .
$K_{j,t}$	Contribution base for policy j in year t . It equals P_t for EIS, unemployment, and old-age insurance, and equals female-worker payroll for maternity insurance.
g_N	Annual covered-worker growth rate.
g_W	Annual wage-growth rate.
i	Annual fund return applied to the opening reserve.
a	Administrative, contingency, and reserve margin applied to base cost.
$Q_{j,t}$	Expected number of claims or beneficiaries for policy j in year t .
$B_{j,t}$	Base benefit cost before the administrative margin.

$F_{j,t}$	Final annual cost after the administrative margin.
c_j	User-entered contribution or premium rate for policy j .
$C_{j,t}$	Contribution income for policy j in year t .
$A_{j,t}$	Accumulated reserve balance for policy j at the end of year t .
r_j^*	Required level premium rate calculated over the projection horizon.

2 Common Projection Framework

All four policies use the same annual population, wage, payroll, loading, contribution, reserve, and premium-rate framework.

2.1 Covered workers

The covered worker population grows geometrically:

$$N_t = N_0(1 + g_N)^t. \quad (1)$$

Thus, each future year's claim exposure changes with the assumed growth or decline of the covered workforce.

2.2 Average monthly wage

The average monthly wage also grows geometrically:

$$W_t = W_0(1 + g_W)^t. \quad (2)$$

Wage growth increases payroll and the wage-linked benefit components. The unemployment service-cost component is different: its fixed per-claim amount does not automatically increase with W_t .

2.3 Annual covered payroll

The model converts monthly average wages into annual payroll:

$$P_t = 12N_tW_t. \quad (3)$$

For three branches, this total payroll is also the contribution base. Maternity uses female-worker payroll as its contribution base. Define the policy-specific base as:

$$K_{j,t} = \begin{cases} P_t, & j \in \{E, U, P\}, \\ 12N_t f W_t = f P_t, & j = M. \end{cases} \quad (4)$$

2.4 Final annual policy cost

Each policy first calculates a base benefit cost $B_{j,t}$. The common administrative margin is then applied:

$$F_{j,t} = B_{j,t}(1 + a). \quad (5)$$

For example, if $a = 15\%$, a base cost of Tk 100 becomes:

$$\text{Tk } 100 \times (1 + 0.15) = \text{Tk } 115. \quad (6)$$

2.5 Contribution income

For policy j , the user enters a contribution rate c_j applied to that policy's contribution base. Annual contribution income is:

$$C_{j,t} = c_j K_{j,t}. \quad (7)$$

The contribution rate is held constant across all projection years unless the user changes the input and recalculates.

2.6 Accumulated reserve

The calculator starts each policy with no initial reserve:

$$A_{j,-1} = 0. \quad (8)$$

The annual reserve recurrence is:

$$A_{j,t} = A_{j,t-1}(1 + i) + C_{j,t} - F_{j,t}. \quad (9)$$

This ordering means:

1. the previous year's closing reserve earns the fund return i ;
2. current-year contributions are added; and
3. current-year costs are deducted.

Current-year contributions do not earn a partial-year return in the formula. Similarly, current-year costs are not discounted within the year.

Expanding the recurrence gives the terminal reserve:

$$A_{j,T-1} = \sum_{t=0}^{T-1} (C_{j,t} - F_{j,t}) (1 + i)^{T-1-t}. \quad (10)$$

2.7 Cumulative cost and contributions

The projection function calculates simple nominal cumulative totals:

$$\text{CumCost}_{j,t} = \sum_{s=0}^t F_{j,s}, \quad (11)$$

$$\text{CumContribution}_{j,t} = \sum_{s=0}^t C_{j,s}. \quad (12)$$

The current interface displays cumulative cost; the cumulative contribution total is also calculated and retained in the projection rows but is not currently displayed. Neither cumulative figure is investment-adjusted. Investment return appears only in the accumulated reserve calculation.

2.8 Required premium rate

The calculator determines a constant contribution-base rate that would make the terminal reserve equal to zero before rounding. It first forms future-value weights:

$$v_t = (1 + i)^{T-1-t}. \quad (13)$$

The unrounded required rate is:

$$\tilde{r}_j^* = \frac{\sum_{t=0}^{T-1} F_{j,t} v_t}{\sum_{t=0}^{T-1} K_{j,t} v_t}. \quad (14)$$

This is the ratio of the terminal-year future value of projected costs to the terminal-year future value of the applicable contribution base. For maternity, the denominator uses female-worker payroll rather than total payroll.

The JavaScript rounds the rate **upward** to four decimal places in decimal form:

$$r_j^* = \frac{\lceil 10,000 \tilde{r}_j^* \rceil}{10,000}. \quad (15)$$

Because the website displays percentages with two decimal places, this is effectively upward rounding to the next 0.01 percentage point.

Actuarial interpretation

The required rate is a level contribution-base rate for the selected horizon and assumptions. It is designed around the final accumulated balance, not a separate annual solvency condition. After upward rounding, applying this rate to the same contribution base produces a terminal reserve that is mathematically non-negative, subject to ordinary floating-point implementation precision. Therefore, it should be read as a scenario rate for this model, not automatically as a statutory or long-term actuarial contribution rate.

3 The Four Policy Calculations

3.1 Employment Injury Insurance

Policy description

The webpage describes a national employment injury scheme intended to replace fragmented employer liability with a no-fault, employer-financed, risk-pooled arrangement. The policy scope includes medical care, temporary incapacity, permanent disability, survivors' pensions, rehabilitation, and return-to-work support. In the calculator, these diverse liabilities are compressed into one average cost per claim measured in months of average wage.

Let:

- e = annual injury claim rate as a proportion of covered workers;
- m_E = average benefit cost in months of average wage.

Expected injury claims are:

$$Q_{E,t} = N_t e. \quad (16)$$

The average cost per claim is:

$$\bar{b}_{E,t} = W_t m_E. \quad (17)$$

The base annual employment injury cost is:

$$B_{E,t} = Q_{E,t} W_t m_E. \quad (18)$$

After the common margin:

$$F_{E,t} = Q_{E,t} W_t m_E (1 + a). \quad (19)$$

Contribution income is:

$$C_{E,t} = c_E P_t. \quad (20)$$

Actuarial interpretation

The parameter m_E is not the duration of one specific payment. It is an aggregate severity measure: the model assumes that the average full cost of one injury claim is equivalent to m_E months of the average wage. A larger value can represent more severe injuries, longer payment duration, pensions, medical costs, rehabilitation, or a combination of these, but the model does not separate those components.

Default first-year illustration

Using the website defaults $N_0 = 1,000,000$, $W_0 = \text{Tk } 15,000$, $e = 0.12\% = 0.0012$, $m_E = 48$, and $a = 15\%$:

$$Q_{E,0} = 1,000,000 \times 0.0012 = 1,200, \quad (21)$$

$$B_{E,0} = 1,200 \times 15,000 \times 48 = \text{Tk } 864,000,000, \quad (22)$$

$$F_{E,0} = 864,000,000 \times 1.15 = \text{Tk } 993,600,000. \quad (23)$$

3.2 Maternity Benefit Insurance

Policy description

The webpage describes a pooled maternity insurance branch intended to protect women workers while spreading maternity cost across covered female-worker payroll. The calculator estimates the number of maternity claims from the female share of workers and a birth/claim rate among female workers. It then calculates wage replacement for a selected number of leave weeks.

Let:

- f = female share of covered workers;
- b = annual maternity claim rate among covered female workers;
- L = paid maternity leave duration in weeks;

- r_M = wage-replacement proportion during leave.

The female worker exposure is $N_t f$. Expected maternity claims are:

$$Q_{M,t} = N_t f b. \quad (24)$$

The code converts weeks to months using 4.345 weeks per month:

$$D_M = \frac{L}{4.345}. \quad (25)$$

The benefit cost per claim is:

$$\bar{b}_{M,t} = W_t \left(\frac{L}{4.345} \right) r_M. \quad (26)$$

The base annual maternity cost is:

$$B_{M,t} = Q_{M,t} W_t \left(\frac{L}{4.345} \right) r_M. \quad (27)$$

After the common margin:

$$F_{M,t} = Q_{M,t} W_t \left(\frac{L}{4.345} \right) r_M (1 + a). \quad (28)$$

The executable calculation uses female-worker payroll for contribution income. Define:

$$K_{M,t} = 12 N_t f W_t = f P_t, \quad C_{M,t} = c_M K_{M,t}. \quad (29)$$

Important implementation note

The current calculator labels the maternity input “Premium, % female-worker payroll.” The JavaScript applies the rate to $12 N_t f W_t$, so male workers are not included in this maternity contribution base. The required maternity rate uses the same female-worker payroll denominator.

Actuarial interpretation

This model separates the claim frequency from the wage-replacement design. The number of claims depends on $N_t f b$, while the cost of each claim depends on wage, leave duration, and replacement rate. It does not separately include medical benefits, antenatal or postnatal service costs, multiple births, eligibility rules, or variation in individual wages.

Default first-year illustration

Using $N_0 = 1,000,000$, $f = 45\%$, $b = 5.5\%$, $W_0 = \text{Tk } 15,000$, $L = 17.14$ weeks, $r_M = 100\%$, and $a = 15\%$:

$$Q_{M,0} = 1,000,000 \times 0.45 \times 0.055 = 24,750, \quad (30)$$

$$B_{M,0} = 24,750 \times 15,000 \times \frac{17.14}{4.345} \times 1.00 \quad (31)$$

$$\approx \text{Tk } 1,464,493,671, \quad (32)$$

$$F_{M,0} \approx 1,464,493,671 \times 1.15 \quad (33)$$

$$\approx \text{Tk } 1,684,167,722. \quad (34)$$

3.3 Unemployment Insurance and Employment Services

Policy description

The webpage describes temporary income protection linked to job search, training, case management, and re-employment. The calculator therefore combines a wage-replacement cash benefit with a fixed employment-service cost per approved claim.

Let:

- u = annual unemployment claim rate as a proportion of covered workers;
- d_U = number of months of cash benefit;
- r_U = wage-replacement proportion;
- S_U = employment-service cost per claim in Taka.

Expected unemployment claims are:

$$Q_{U,t} = N_t u. \quad (35)$$

The cash benefit per claim is:

$$\bar{b}_{U,t}^{\text{cash}} = W_t d_U r_U. \quad (36)$$

The total modeled cost per claim is:

$$\bar{b}_{U,t} = W_t d_U r_U + S_U. \quad (37)$$

The base annual unemployment cost is:

$$B_{U,t} = Q_{U,t} (W_t d_U r_U + S_U). \quad (38)$$

After the common margin:

$$F_{U,t} = Q_{U,t} (W_t d_U r_U + S_U) (1 + a). \quad (39)$$

Contribution income is based on total covered payroll:

$$C_{U,t} = c_U K_{U,t} = c_U P_t. \quad (40)$$

Actuarial interpretation

The fixed service cost S_U grows with the number of claims but does not grow automatically with wages. By contrast, the cash-benefit component grows with W_t . The administrative margin is applied to both the cash benefit and the service cost because both are included in the base cost before loading.

Default first-year illustration

Using $N_0 = 1,000,000$, $u = 3.5\%$, $W_0 = \text{Tk } 15,000$, $d_U = 4$, $r_U = 50\%$, $S_U = \text{Tk } 8,000$, and $a = 15\%$:

$$Q_{U,0} = 1,000,000 \times 0.035 = 35,000, \quad (41)$$

$$\bar{b}_{U,0} = 15,000 \times 4 \times 0.50 + 8,000 = \text{Tk } 38,000, \quad (42)$$

$$B_{U,0} = 35,000 \times 38,000 = \text{Tk } 1,330,000,000, \quad (43)$$

$$F_{U,0} = 1,330,000,000 \times 1.15 = \text{Tk } 1,529,500,000. \quad (44)$$

3.4 Old-Age Social Insurance

Policy description

The webpage describes a contributory old-age pillar that complements tax-financed old-age assistance. In the calculator, annual pension expenditure is approximated from the number of retirees relative to active workers and a pension replacement rate applied to the average annual wage.

Let:

- ρ = retiree-to-active-worker ratio;
- r_P = pension replacement proportion of average wage.

The number of pensioners supported is approximated as:

$$Q_{P,t} = N_t \rho. \quad (45)$$

The annual pension per retiree is:

$$\bar{b}_{P,t} = 12W_t r_P. \quad (46)$$

The base annual pension cost is:

$$B_{P,t} = Q_{P,t} 12W_t r_P. \quad (47)$$

After the common margin:

$$F_{P,t} = Q_{P,t} 12W_t r_P (1 + a). \quad (48)$$

Contribution income is based on total covered payroll:

$$C_{P,t} = c_P K_{P,t} = c_P P_t. \quad (49)$$

Because $Q_{P,t} = N_t \rho$ and $P_t = 12N_t W_t$, the base pension cost can be simplified to:

$$B_{P,t} = P_t \rho r_P. \quad (50)$$

Therefore, with constant ρ , r_P , and a , the loaded annual pension-cost rate is:

$$\frac{F_{P,t}}{P_t} = \rho r_P (1 + a). \quad (51)$$

Actuarial interpretation

This is an aggregate pay-as-you-go-style expenditure approximation. It is not an individual-account accumulation model and does not calculate the present value of accrued pension rights. It has no retirement-age distribution, mortality table, contribution history, vesting period, survivor pension, indexation rule, or life expectancy assumption.

Default first-year illustration

Using $N_0 = 1,000,000$, $\rho = 6\%$, $W_0 = \text{Tk } 15,000$, $r_P = 25\%$, and $a = 15\%$:

$$Q_{P,0} = 1,000,000 \times 0.06 = 60,000, \quad (52)$$

$$\bar{b}_{P,0} = 12 \times 15,000 \times 0.25 = \text{Tk } 45,000, \quad (53)$$

$$B_{P,0} = 60,000 \times 45,000 = \text{Tk } 2,700,000,000, \quad (54)$$

$$F_{P,0} = 2,700,000,000 \times 1.15 = \text{Tk } 3,105,000,000. \quad (55)$$

4 Meaning of the Calculator Outputs

4.1 Claim rate

The displayed claim rate is the user-entered policy parameter:

- EIS: injury claims per covered worker;
- maternity: claims per covered female worker;
- unemployment: unemployment claims per covered worker;
- old age: retirees per active covered worker.

For maternity, the displayed birth/claim rate is not the claim rate for all workers. Total claims are calculated only after multiplying by the female share.

4.2 Final-year claims

This is $Q_{j,T-1}$: the expected claims or beneficiaries in the last projection year after worker growth has been applied.

4.3 Final-year cost

This is $F_{j,T-1}$: the last year's base benefit cost multiplied by $(1 + a)$.

4.4 Required premium rate

This is r_j^* , the upward-rounded level rate on the applicable contribution base that, when applied to projected contribution bases, balances the future value of projected costs at the end of the selected horizon. Maternity uses female-worker payroll; the other three branches use total covered payroll.

4.5 User-set contribution rate

This is c_j , the rate entered in the webpage. The calculator compares it with r_j^* and marks the policy as underfunded when:

$$c_j < r_j^*. \quad (56)$$

4.6 Accumulated amount

This is the closing reserve $A_{j,t}$. A negative value means projected costs and prior deficits exceed contributions and investment earnings under the selected assumptions. A positive value means an accumulated surplus remains.

5 What Is and Is Not Actuarially Modelled

5.1 What the model does capture

- exposure growth through covered workers;
- wage-linked payroll and benefit components, plus the fixed nominal unemployment service-cost component;
- policy-specific claim frequency or beneficiary ratios;
- policy-specific benefit severity or replacement parameters;
- an administrative/contingency loading;
- contribution income;
- annual reserve accumulation with investment return; and
- a horizon-based required level premium rate.

5.2 Main simplifications

The webpage itself describes the calculator as simplified. Mathematically, the following elements are not represented:

- stochastic variation or confidence intervals;
- age, sex, wage-band, occupation, sector, or service-duration distributions;
- changing claim rates over time;
- mortality, morbidity, disability severity, recovery, or remarriage assumptions;
- waiting periods, eligibility conditions, contribution-density requirements, or repeat claims;
- benefit caps, floors, indexation, survivor proportions, or commutation factors;
- separate medical, rehabilitation, administration, and cash-benefit cost streams;
- initial reserves or legacy liabilities;
- separate investment timing for contributions and benefit payments;
- inflation distinct from wage growth; and
- policy interaction, covariance, or cross-subsidy among the four branches.

Correct interpretation

The website is best understood as a transparent policy-scenario calculator. It answers: “Given these assumed claim rates, benefit designs, payroll growth, loading, contribution rates, and investment return, what annual costs and reserve path result?” It does not by itself establish that the assumptions are statistically credible or that the resulting rate is adequate for a permanent national scheme.

6 Compact Formula Summary

Component	Formula
Workers	$N_t = N_0(1 + g_N)^t$
Wage	$W_t = W_0(1 + g_W)^t$
Payroll	$P_t = 12N_tW_t$
Contribution base	$K_{j,t} = P_t$ for $j \in \{E, U, P\}$; $K_{M,t} = 12N_tfW_t = fP_t$
EIS claims	$Q_{E,t} = N_te$
EIS base cost	$B_{E,t} = Q_{E,t}W_tm_E$
Maternity claims	$Q_{M,t} = N_tfb$
Maternity base cost	$B_{M,t} = Q_{M,t}W_t(L/4.345)r_M$
Unemployment claims	$Q_{U,t} = N_tu$
Unemployment base cost	$B_{U,t} = Q_{U,t}(W_td_Ur_U + S_U)$
Pensioners	$Q_{P,t} = N_t\rho$
Old-age base cost	$B_{P,t} = Q_{P,t}W_t12r_P$
Loaded final cost	$F_{j,t} = B_{j,t}(1 + a)$
Contribution income	$C_{j,t} = c_jK_{j,t}$
Reserve recurrence	$A_{j,t} = A_{j,t-1}(1 + i) + C_{j,t} - F_{j,t}$
Raw required rate	$\tilde{r}_j^* = \frac{\sum_t F_{j,t}(1 + i)^{T-1-t}}{\sum_t K_{j,t}(1 + i)^{T-1-t}}$
Displayed required rate	$r_j^* = \lceil 10,000\tilde{r}_j^* \rceil / 10,000$

7 Source Mapping Within the Website

This explanation is relevant to the developed website.

Website location	Content used
<code>index.html</code> , policy-model section	Policy goals, coverage, benefits, and financing descriptions for the four branches.
<code>index.html</code> , calculator panels	Default input values and user-facing parameter labels.
<code>index.html</code> , definitions section	Plain-language meanings of inputs and outputs.
<code>script.js</code> , input-reading block	Percentage conversion, bounds, and parameter names.
<code>script.js</code> , policy array	Claim-count and base-cost formula for each policy.
<code>script.js</code> , projection function	Worker and wage growth, payroll, policy-specific contribution bases, margin, contributions, reserve recurrence, cumulative totals, and required-rate calculation.